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Using social media and citizen science to assess cultural ecosystem services along riverine landscapes

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Abstract

Rivers offer cultural ecosystem services (CES) that improve people's quality of life. Advancements in computing and data storage have primarily focused on terrestrial CES, neglecting riverine areas. This study aims to develop a methodology to assess CES in riverine landscapes from social media and citizen science images related to environmental information. We collected georeferenced pictures from Flickr and iNaturalist for three main test rivers in northwest Portugal (Minho, Lima and Cávado) and classified them based on content such as 'biodiversity', 'recreation/river beaches', 'historical heritage' and 'landscape', as well as environmental spatial variables. A multimodel inference approach was applied to predict the spatial distribution of the pictures and environmental variables to support CES mapping. The methodology was applied during two time periods, before and during the most restrictive period of the COVID-19 pandemic. Results showed that estuaries were identified as 'hotspots' for CES related to rivers provision. There was distinct prevalence of pictures depending on the targeted river: pictures exhibiting 'recreation/river beaches' prevailed in Cávado (62%), 'biodiversity' in Lima (70%) and 'historical heritage' in Minho (39%). Only the values and patterns from the category 'biodiversity' were maintained on the two analysed periods, with the other categories not having posts in social media during COVID-19 most restrictive period. The methodology for CES assessment in rivers can be replicated using different time periods and regions due to its simple stepwise framework. The study provides valuable insights for sociocultural approaches, aiding in decision-making on freshwater environment management, despite potential limitations in image distribution.

KEYWORDS
cultural ecosystem services (CES), multimodel inference, mapping river CES, social media, NW Portugal, COVID-19

1 | INTRODUCTION

Cultural ecosystem services (CES) provide numerous benefits to people's quality of life, either through physical (e.g., recreation and ecotourism) or intellectual (e.g., historical heritage) interactions (Boulton et al., 2016; Carucci et al., 2022; Fish et al., 2016). Human–nature interactions, which include many of CES, have raised special attention from the media and research community through the general lockdowns during the most critical periods of the COVID-19 pandemic (Beckmann-Wübbelt et al., 2021; Martinez-Harms et al., 2018). However, relatively to other ecosystem services types, such as provisioning or regulating, the assessment of CES has for long been rather limited (Crouzat et al., 2022; Rendon et al., 2019; Veerkamp et al., 2021).

Conventional methods addressing CES include the use of participatory approaches, such as social-based questionnaires (Wolff et al., 2015), which can be high-demanding when reaching extensive audiences as well as spatial and temporal coverages (Yoshimura & Hiura, 2017). Complementary approaches to address CES can be found in emergent fields that harness the use of online digital data to comprehend human–nature interactions, including digital conservation (Arts et al., 2015), Conservation Culturomics (Ladle et al., 2016) or even iEcology (Jarić et al., 2020).

Among the relevant sources of online digital data to address CES are social media and citizen science platforms, such as Flickr, iNaturalist or Wikiloc (Cardoso et al., 2022; Rosário et al., 2019; Vaz et al., 2020; Zhang et al., 2020). Such platforms hold imagery information, that is, pictures taken and shared by which contain a variety of nature-related content. Alongside the content, online images are often linked to textual information, such as captions, descriptions or tags, which content can also be an added-value to comprehend multiple CES dimensions (Heikinheimo et al., 2017; Pastur et al., 2016). Furthermore, online digital data often include time stamps and geo tags, which makes it quite useful to understand when and where imagery or textual content have been created and hence get insights into CES availability and location (Clemente et al., 2019; Majid et al., 2013). In fact, the biophysical context is also crucial for mapping and valuing CES in the context of ecosystems that provide these services (Van Berkel et al., 2018; Vaz et al., 2019; Vaz et al., 2022). Geographic Information Systems (GIS) for biophysical data management can be very useful in describing and analysing the biophysical context in which each photograph is georeferenced on social networks (Xu et al., 2020) and analysing the biophysical context and attributes of nature that support CES opportunities, for example by informing about the accessibility of resources or understanding visual-sensory features present in the landscape or the topography of a particular location (Tveit et al., 2006; Vaz et al., 2022). Thus, when combined, we can enjoy a very promising tool for monitoring CES in rivers.

Alongside the biophysical context, CES are also tightly linked with the way people interact with nature and hence allow the co-creation of physical or mental benefits, for example, though nature walking, outdoor social events, among many others (Zhang et al., 2020). As such, social media content can help to understand people's interests and interactions in the wild, for example, by interpreting images posted by people when enjoying a walk in a nature trail or taking a selfie with a peculiar natural landscape in the background (Crouzat et al., 2022). Nonetheless, social media content also holds limitations when addressing CES (Cheng et al., 2019). The methodologies produce a huge amount of data, which often requires substantial filtering processes to identify 'valid' data, and often this process has to be done manually (Hale et al., 2019). On the other hand, there is a problem with the creation of ghost users who cause the creation of content on social networks that is not published by a person but by the algorithm of a computer (Ismail & Latif, 2013). Finally, there is still a very large inequality in obtaining the internet and consequently in the use of social media, so we always have a dataset influenced only by people who have access to the internet (Martinez-Harms et al., 2018).

Getting meaningful information for CES requires dealing with the vast amount of digital data created and shared by social media users on a daily basis, accessing can be time-consuming particularly when done manually (Hale et al., 2019). A way to overcome this is to rely on artificial intelligence models, for example, deep learning (Cardoso et al., 2022; Christin et al., 2019), which although helpful to address big social media data still pose additional challenges, the availability of computer facilities and open access code (Campbell et al., 2013; Peterson et al., 2010) or even the maintenance of digital infrastructures and data maintenance protocols (Peterson et al., 2010).

Using social media to address CES has been well established in terrestrial ecosystems; however, it has not been applied so often to freshwater ecosystems, in particular to rivers (Tulek, 2023). River ecosystems provide cultural activities, such as landscape appreciation, religious sanctuaries and cultural festivals that benefit humans (Dai et al., 2023; Thiele et al., 2019; Wantzen et al., 2016). Riverside areas also offer recreational activities, such as canoeing, cycling, sport fishing and hiking, which significantly improve citizens' quality of life and local economies (Hale et al., 2019; Thiele et al., 2020). Studies that address the CES of rivers include the valorisation of various themes such as landscape, historical, artistic and cultural expression, religious, spiritual or ceremonial activities, environmental research and teaching and, finally, recreational activities as the most prevalent categories along rivers (Hale et al., 2019; Thiele et al., 2019; Thiele et al., 2020; Tulek, 2023). Still, to our knowledge, the utility of social media to address CES in river ecosystems is far from being explored.

The aim of this study was to develop a methodology to model and map CES related to rivers through photographic data from social media and citizen science platforms, together with spatial variables to support management and conservation measures in river ecosystems. Taking three main rivers from northwest Portugal (Minho, Lima and Cávado) as test areas, we specifically aim to clarify three main questions: (i) Which categories of CES do rivers and riparian areas provide for human well-being based on the content of social media citizen science platforms pictures? (ii) Which are the spatial and environmental variables that best explain the distribution of social media photographs and the demand for CES in the study areas? and (iii) Is the mapping of such social media pictures useful to infer possible hotspots of CES interest along riverine areas? Our methodology is tested for two time periods, coinciding with typical and atypical years, January 2018 to December 2019 and during the most restrictive period of COVID-19 Pandemic, March 2020 to March 2021.

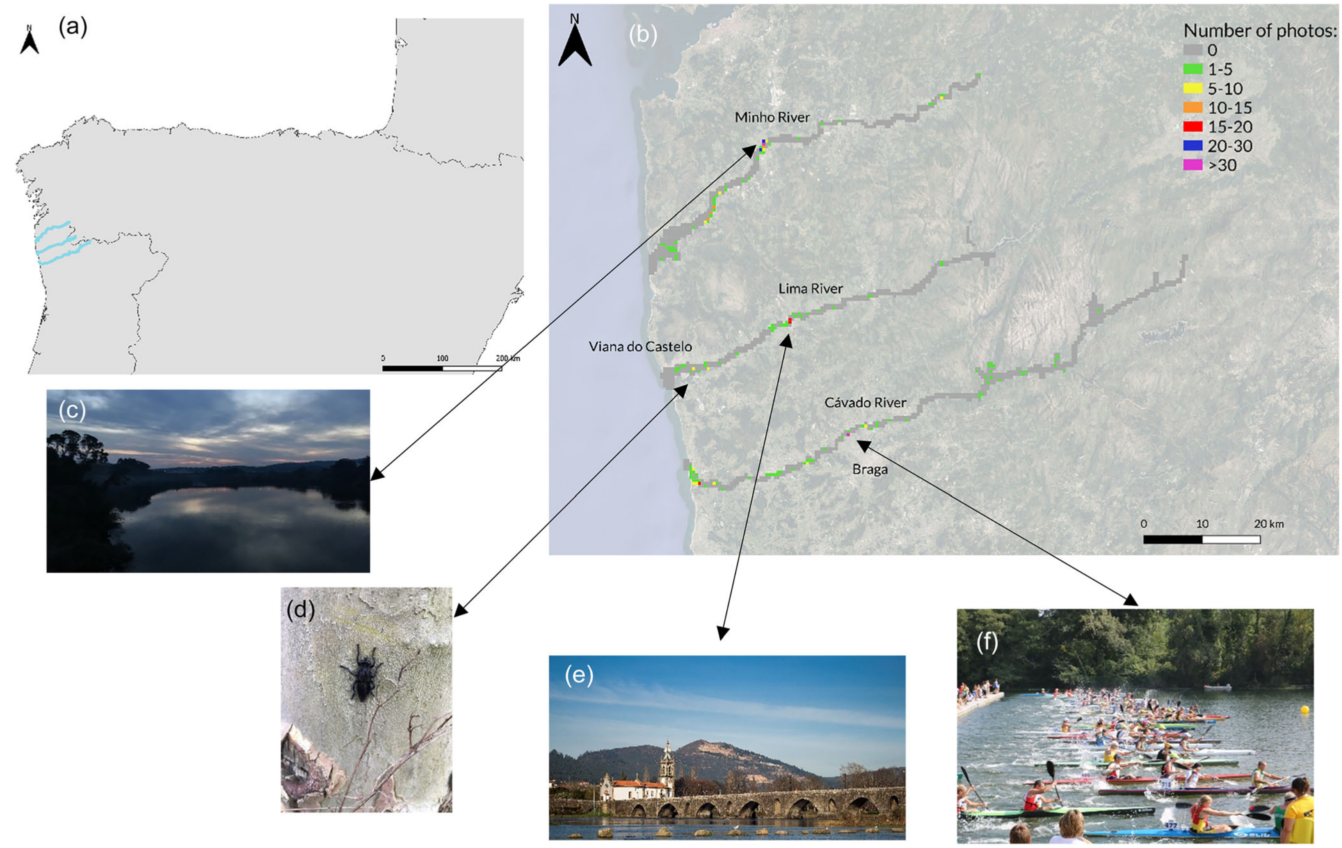

FIGURE 1 (a) Location of the study area and Minho, Lima and Cávado rivers; (b) distribution of the number of photographs by the study area with 4 photos representing the categories: (c) Landscape (Source: Flickr, Sergei Gussev), (d) Historical heritage (Source: Flickr, Richard Alonso), (e) Biodiversity (Source: iNaturalist, miguelborges), (f) Recreation/river beaches (Source: Flickr, jmquintanamurillo).

2 | METHODS

2.1 | Case Study

The study area focused on three main rivers of northwest Portugal, namely, Minho, Lima and Cávado rivers, all draining to the Atlantic Ocean (Figure 1).

The Minho River is a significant river that flows along the northern border of Portugal with Spain, started in the Serra do Meira in Spain and meanders south-westward for approximately 315 km before entering into the Atlantic Ocean (Pacheco, 2014) . The Lima River is also an international river, with origin in the Serra of São Mamede and flows for approximately 137 km before reaching the Atlantic Ocean (Costa et al., 2017) . Finally, the Cávado River is fully Portuguese, originating in the Serra do Larouco, and flows for approximately 135 km, encountering a vast cascade of 9 dams, before entering Esposende the estuary (Lacasta & Seixas, 2016). The three rivers are located in the Minho region, which has been an important touristic attraction over the last years, known for its lush green landscapes, historic towns and 'verde wine'. These areas are particularly important in terms of conservation, as they are mentioned in the lists of sites protected by the Natura 2000 Network and Protected Areas associated with species listed in the European Union Habitats and Birds directives. For example, the area of the Peneda-Gerês National Park, which encompasses the Portuguese headwaters of the three rivers, has a variety of protected habitats such as the wet moors of *Erica ciliaris* (Fernandes, 2008a). It also features special protection zones (ZPE) as they constitute favourable habitat for a great diversity of species, highlighting the priority habitats of alluvial forest of willows and alders (habitat 91E0) (Fernandes, 2008b). The Minho and Lima river estuaries are considered a special zone for conservation (ZEC) (Da Silva, 2012; Lacasta & Seixas, 2016). In addition, Cávado River estuary is included in a protected area Litoral Norte Natural Park encompassing 16 km of coastline between the mouth of the river and the Apúlia area, in the municipality of Esposende. This Litoral Norte Natural Park has, according to Natura 2000 Network, the status of Site of Community Importance (SCI) and is also a protected area (Lacasta & Seixas, 2016). According to the latest census in 2021, Braga is the municipality with the highest population density, with 192,000 inhabitants located in the Cávado basin, followed by Viana do Castelo with 37,000 inhabitants located in the Lima basin, and the less populated is the Minho River basin, with the municipality of Valença as the largest city in the basin with 14,000 inhabitants (Instituto Nacional de Estatística, 2021). In recent studies for CES in the Peneda-Gerês National Park, people's accessibility to leisure elements and landscape visual-sensory

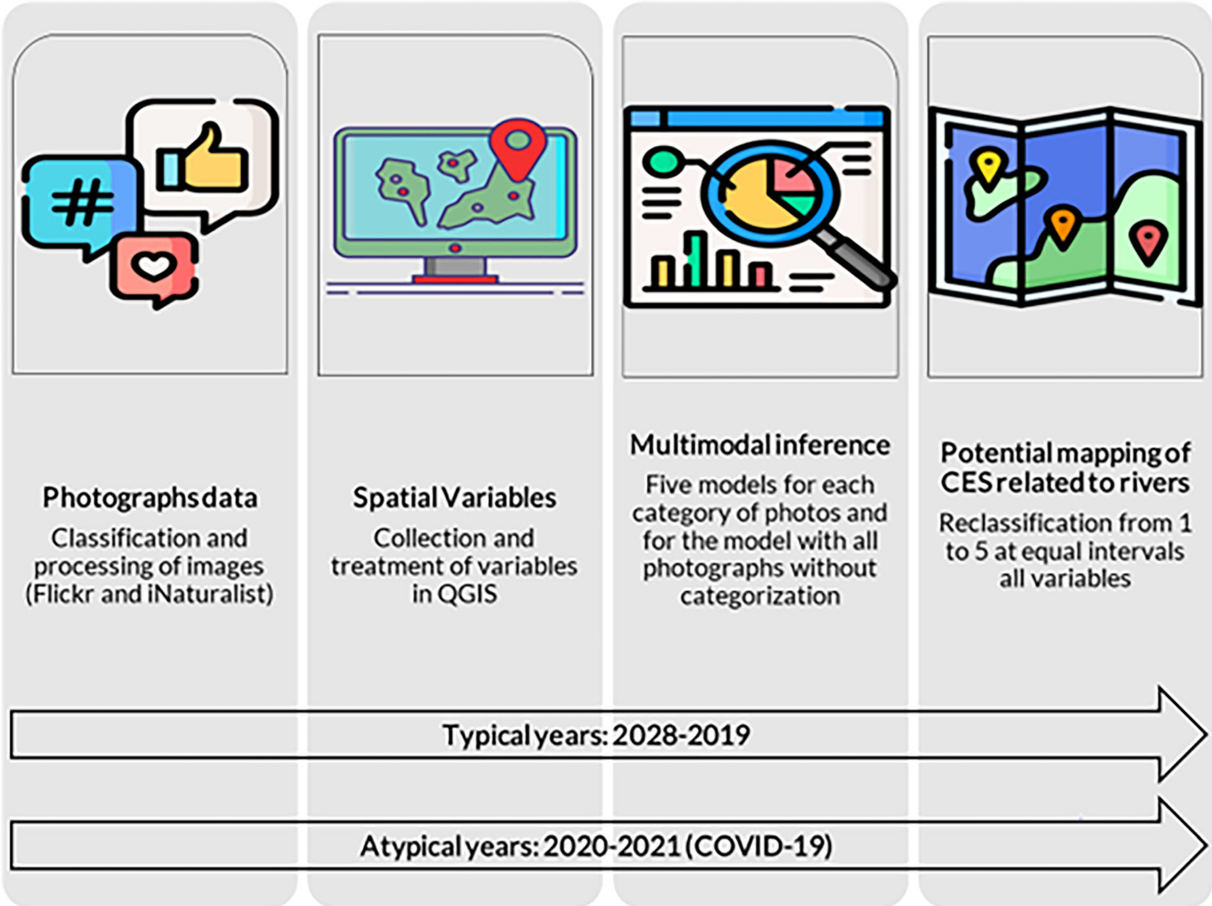

FIGURE 2 Methodological framework adopted to study cultural ecosystem services (CES) related to rivers.

characteristics were identified as the main spatial variables explaining the demand for CES (Cardoso et al., 2022; Crouzat et al., 2022).

2.2 | Methodological framework

The proposed steps provide a transferable approach (Figure 2) based on social network data to detect the relationship between river CES and their respective biophysical contexts.

The first step was to download photos related to the respective rivers under study from the social media and a citizen science platform (Flickr and iNaturalist, respectively) for two different time periods: representative of typical years (between 2018–2019); and representative of atypical years (march 2020 to march 2021) corresponding to the most restrictive period of the COVID-19 pandemic. The split analysis allowed the evaluation of the effectiveness on mapping potential location for CES if the photo sample is different (Section 2.3). Still in this step and after downloading, we manually categorized the photos into four categories linked to the river CES: (i) landscape, (ii) historical heritage, (iii) biodiversity and (iv) recreation/river beaches (Table 1), for the two time periods. Only these CES categories related to rivers were chosen because they are the only ones represented in the photographic data collected on the social networks with which we work.

Secondly, a set of environmental and spatial explanatory variables were compiled that justified the presence of CES-related photographs and were mainly related to the environmental context of the areas (Section 2.4).

Third, multimodel inference was applied as statistical modelling approach. Multimodel inference serves to assess the explanatory power of the variables as a function of presence of CES represented as content of social networks posts (Section 2.5).

Finally, the potential mapping along the three rivers was performed through the selected variables from the multimodel inference (section 2.6).

2.3 | Photographs data

The photographs to evaluate CES related to the rivers were taken from Flickr (https://www.flickr.com/) (Hale et al., 2019; Ghermandi et al., 2020; Cardoso et al., 2022) and iNaturalist (https://www.inaturalist.org/) (Belaire et al., 2015; Havinga et al., 2020) corresponding to the two periods mentioned in the previous section. We chose to use these two time periods in order to understand

TABLE 1 Categories of CES used in the content analysis of Flickr and iNaturalist photographs according to the Common Ecosystem Services Classification (CICES) (Haines-Young & Potschin, 2018).

Category of cultural ecosystem services	Description
i) Landscape	These are photographs in which the user highlights elements of nature and landscape in a horizontal plane
ii) Historical heritage	These are photographs showing historical monuments linked to culture and religion (e.g., churches, bridges and walkways).
iii) Biodiversity	In these photographs, fauna and/or flora are highlighted as the main theme (e.g., close-up of species)
iv) Recreation/river beaches	These are photographs where the user highlights recreational activities that are closely linked to river beaches (e.g., sunbathing, swimming, sport fishing or water sports).

Abbreviation: CES, cultural ecosystem services.

whether there were changes in the demand for CES illustrated by social media users. A buffer of 500 m around the main channel of the three rivers was made, and only the photographs geographically located within the buffer were selected. Subsequently, each photograph was manually classified into four categories according to the Common Ecosystem Services Classification (CICES) (Haines-Young & Potschin, 2018) as follows: (i) landscape, (ii) historical heritage, (iii) biodiversity and (iv) recreation/river beaches (Table 1). In the end, we obtained 1125 photographs from the Flickr platform and 311 photographs from the iNaturalist platform: there was a final set of 1048, 98 and 290 photographs corresponding to the Cávado, Lima and Minho rivers, respectively. In the year 2020–2021, we obtained 5 photos from Flickr and 307 photos from iNaturalist: there was a final set of 165, 101 and 46 photographs corresponding to the Cávado, Lima and Minho rivers, respectively.

2.4 | Spatial variables

We collected 11 candidate environmental spatial variables (Appendix A1, Appendix S1) related to the biophysical context to model and map the CES related to rivers. The variables are land cover classes, recreation and tourism points (points of interest on the map, e.g., canoeing school), accessibility (distance from tourist spots to urban areas), slope, average annual precipitation, average annual temperature, number of tracks, tracks popularity (number of times a track was used), number of river beaches (drawn by satellite images on Google Earth), viewshed dimension (how many points within a 5-km radius can be seen) and protected areas (existence or not). A regular grid of 500 $\times$ 500 m was established to aggregate by pixel grid all spatial variables, using QGIS 3.16.11 software and ranking them by equal interval classification (Appendix B1). Some of these variables such as land cover type, accessibility to urban areas, viewshed dimension, number of trails and number of recreation and tourism points were identified in previous studies in the region as relevant spatial variables to explain the demand for CES as they are statistically explanatory with the presence of photographs (Cardoso et al., 2022).

2.5 | Multimodel inference

For each grid cell, the number of photographs from each category of CES (Table 1) was used as response spatial variables in a multimodel inference (Burnham & Anderson, 2002). Multimodel inference was chosen since best explanatory variables are selected by making inference from all the multiple competing models in the candidate set (Burnham & Anderson, 2002). Models with ΔAICc ≤2 relative to the top-ranked model were retained (Zuur et al., 2010). Five models were considered: (i) one for the landscape category photographs, (ii) one for historical heritage, (iii) one for biodiversity, (iv) recreation/river beaches and (5) one for all photographs. The multimodel performance criteria used were: Akaike information criterion (AIC), weight (wi) and explained adjusted deviation (D2) (Burnham & Anderson, 2002) through the R language (R, C, 2022) of the RStudio software (Rstudio, 2022). To avoid autocorrelation and multicollinearity between the spatial variables, only predictors with a paired Spearman test value of less than 0.6 (Fox & Weisberg, 2018) and with variance inflation factor, VIF > 2 (Zuur et al., 2009) were excluded. Nine final variables were considered for the construction of the models (Appendix A1), and the variables slope and average annual precipitation were excluded from the model.

2.6 | Potential mapping of CES related to rivers

To map potential CES related to rivers, the previous selected spatial variables from multimodel inference were used. Each spatial variable has been reclassified in QGIS 3.16.11 software from 1 to 5 in equal intervals (Appendix B1, Appendix S1). Subsequently, for each best model, the maps were summed and divided arithmetically by the number of variables, by pixel.

3 | RESULTS AND DISCUSSION

3.1 | Photographs from social media and citizen science platforms

A total of 1436 (100%) photographs were collected for the typical period (2018–2019), out of which 1150 photographs (80%) were from the Flickr platform and 286 photographs (20%) from iNaturalist. The most represented category from the evaluation of photographs was 'recreation/river beaches' with 62% gathering 1436 photographs collected (Figure 3), mostly represented in the Cávado River (84%). To

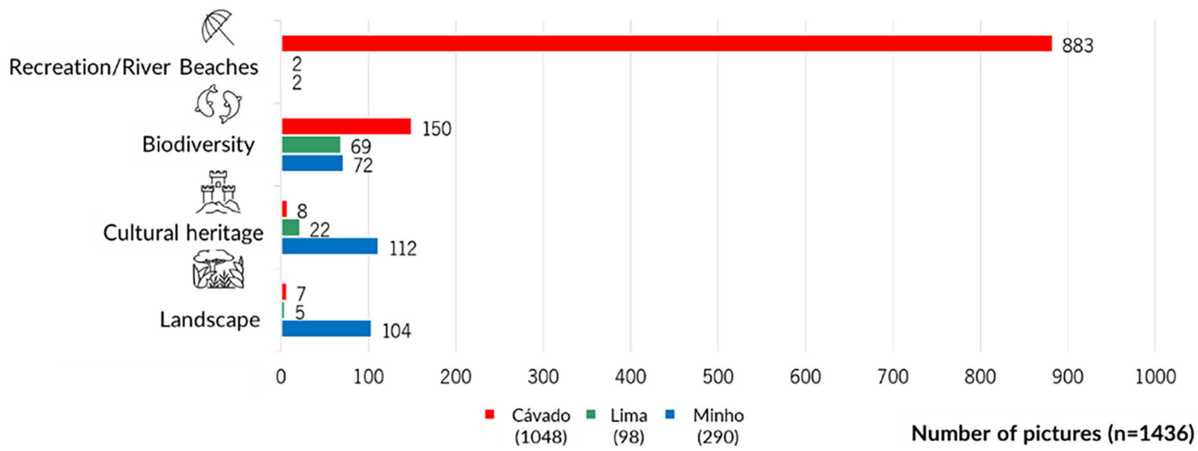

FIGURE 3 Number of photographs from Flickr and iNaturalist of each category of cultural services on the Cávado, Lima and Minho rivers between 2018 and 2019.

note, this is the river with more beaches compared with the other two rivers and excellent facilities and conditions for bathing activities (Oliveira, 2019). Notably, 883 of these photographs from ‘recreation/river beaches’ correspond to a single geographic point coinciding with the river beach of Merelim (São Paio) in the Cávado River, close to the most important city in the region, Braga. In the Lima River, the category most represented in the photographs was biodiversity (70%). This finding is consistent with previous research indicating that protected areas promote a greater appreciation of biodiversity, closely related to CES, specifically spirituality and cultural identity (Haines-Young & Potschin, 2018; Vaz et al., 2020). In fact, the Lima River basin has a vast area included in Natura 2000 network, namely, the Peneda-Gerês National Park, the Bertiandos Lagoons and the ZEC Lima estuary. In the Minho River, the most representative category was ‘cultural heritage’, which is in line with the few monuments or infrastructures near its banks that may entice people to visit, such as the Cathedral of St. Mary of Tui (Spain) and the Valencia Railway Bridge. It is also known that the Minho region has always been closely linked to the catholic religion throughout history (Vaz et al., 2020), which encourages the visits to these monuments. Different categories are more prevalent for each river, indicating that the social media users attribute varying values to different CES in the three river systems. According to Mouttaki et al. (2022) on the coast of Lithuania in the southeastern part of the Baltic Sea, the three most represented categories in the photographs were ‘Landscape Appreciation’, ‘Nature Appreciation’ and ‘Historical Monuments’ (Mouttaki et al., 2022), coinciding to our ‘landscape’, ‘biodiversity’ and ‘historical heritage’, but in our study, the first was not ranked as first category identified. Conversely, Tulek (2023) in the Lower Rhine River basin in the Netherlands pointed ‘Landscape Appreciation’ and ‘Natural Structures’ as the most prominently represented categories, more similar to the category ‘landscape’ in our study that was not ranked as first in none of our rivers.

Finally, it is worth noting that the Flickr platform used in our photographic data retrieval went bankrupt in 2018. In April 2018, the platform's photographs were transferred from the Yahoo company server to the SmugMug company server, resulting in the deletion of thousands of photographs from the platform (Vox, 2019). This situation may have influence in the number of photographs from January to April 2018 that could have been deleted, but since the period of our analysis last until December 2019, we considered to have enough representation, also gathering data from another platform, iNaturalist.

3.2 | Explanatory variables of CES related to rivers

In our study, the explanatory spatial variables that stand out in all CES categories were (i) protected areas, (ii) distance from river beaches, (iii) tourist attractions to urban areas (accessibility), (iv) number of trails, (v) recreation and tourism points and (vi) average annual temperature (Table 2). The model with the lowest ΔAICc (ΔAICc ≤ 2) was chosen for each category of photographs (Appendices C1 and C2). Protected areas are a crucial factor in the explanation of photos, as those who live in such areas are more likely to cherish and preserve them (Ament et al., 2017). Accessibility emerged as one of the best explanatory variables, corroborating findings from other studies that indicate locations with easy access are more likely to employ CES in general (Crouzat et al., 2022). Furthermore, the accessibility of a site correlates with the number of tracks as many sites are reachable not by cars or other means of transportation but rather through walking or biking tracks, fostering enhanced interaction with the available CES (Schamel & Job, 2017). A huge step towards outdoor leisure, hiking and even landscape appreciation is the establishment of a track network (Schamel & Job, 2017). According to Bachi et al. (2020), there is a positive relationship between places with more points of recreation and tourism and the demand for CES. This finding can bring valuable insights to enhance the visitation experiences by formulating comprehensive plans across municipalities, recognizing their potential as areas of high demand for CES demand and,

TABLE 2 Multimodel inference results: Akaike information criterion (AIC), multimodel inference weight (Weight) and adjusted explained variance (D2) of the inference models with the best results of the explanatory variables for each CES category (inferred from the content of social media photographs in the years 2018/2019) and the most restrictive period relative to the COVID-19 pandemic (the variables and values in green are those corresponding to the inference of the models relative to the year relative to the COVID-19 pandemic for comparison purposes).

Best models for each category	AIC	Weight	D2
All photographs			
(AP + APPAU + NT + OS + P + PRT + TMA + VD)	8634	0.44	0.99
(APPAU+NT + OS+P + PRT + PT + TMA + VD)	1646	0.58	0.47
Biodiversity			
(AP + APPAU + NT + OS + PRT + TMA + VD)	1561	0.25	0.47
(APPAU + NT + OS + P + PRT + PT + TMA + VD)	1638	0.57	0.47
Recreation/river beaches			
(AP + APPAU + NT + P + PRT + TMA + VD)	521	1	0.99
Landscape			
(AP + APPAU + NT + OS + PRT + PT + TMA + VD)	634	0.33	0.62
Cultural heritage			
(AP + APPAU + NT + OS + P + PRT + TMA)	420	0.47	0.77
(NT + PRT)	40	0.10	0.08

Abbreviation: CES, cultural ecosystem services.

consequently, facing increased environmental pressure (Schirpke et al., 2016; Bachi et al., 2020). Finally, the viewshed dimension emerged as a very important predictor of CES, as social network users tend to capture landscape photos in places with a wide and unobstructed views, often seeking to appreciate the surrounding scenery reflected in this type of photography (Schamel & Job, 2017).

3.3 | Potential mapping of CES related to rivers

Based on the spatial projections of CES related to rivers (Figures 4 and 5), it is evident that areas closer to the coast, particularly those related to estuaries, have a greater potential to provide CES related to rivers. We found that the great hotspot of cultural services provision is the Minho River estuary. Moreover, the Minho River had the highest potential score compared to Lima and Cávado rivers. In contrast, the headwaters of the Cávado River in Montalegre had the lowest potential for CES related to rivers. As stated in the previous section, accessibility stands out as one of the most important factors in substantiating the appreciation of CES. Indeed, the most remote and distant areas, such as the municipalities of Montalegre, have the lowest potential for CES, also materialized in the low number of shared pictures in social media and citizen science images (Figure 1). This could imply that areas situated far from urban centres might exhibit lower levels of CES supply, although the potential for provision could exist. In addition, there appears to be a lower proclivity among rural local population to use social media platforms. There are still people who do not have smartphones or computers in some cases (Chen et al., 2018) resulting in limited access to digital data sharing platforms, either due to socioeconomic or cultural reasons. As a consequence, these disparities create inequalities in the way CES are used. Generally, people from wealthier municipalities travel more and visit protected areas more frequently, whereas residents from poorer municipalities only visit protected areas near their home places (Martinez-Harms et al., 2018; Shanahan et al., 2014). If this is the case, the most marginalized people not only lack access to platforms and social networks but also miss out the variety of CES available in ecosystems. Striking an appropriate balance between development and conservation, and bridging the gap between natural sciences and social sciences, becomes crucial to promote both the conservation of these CES and ensuring equitable access and utilization for all (Oldekop et al., 2016; Sarkki & Acosta García, 2019).

3.4 | Testing the methodology for the most restrictive period of the COVID-19 pandemic

The patterns for photographic data corresponding to the year with more social restrictions during COVID-19 pandemic (between March 2020 and March 2021) differed significantly from those obtained in the annual average 2018/2019, with exception for 'biodiversity' category (Figure 6). The total number of photographs has decreased from 874 (average of 2 years) to 312. Of these photographs, 2% correspond to the Flickr platform and 98% to iNaturalist. Furthermore, during COVID-19, only two categories were present: 'biodiversity' and 'historical heritage', with the biodiversity category being the most representative, and historical heritage having only 5 pictures posted compared to the 64 from 2018/2019. Considering the map projections of CES (Figures 7 and 8), we can see that the model for all categories including biodiversity continues with the same pattern, in which the estuary areas have greater potential, namely the Minho estuary.

FIGURE 4 Cultural ecosystem services (CES) spatial projection of all categories. The red represents the places with the highest potential for CES provisional and the dark green the places with the lowest potential according to the selected spatial variables.

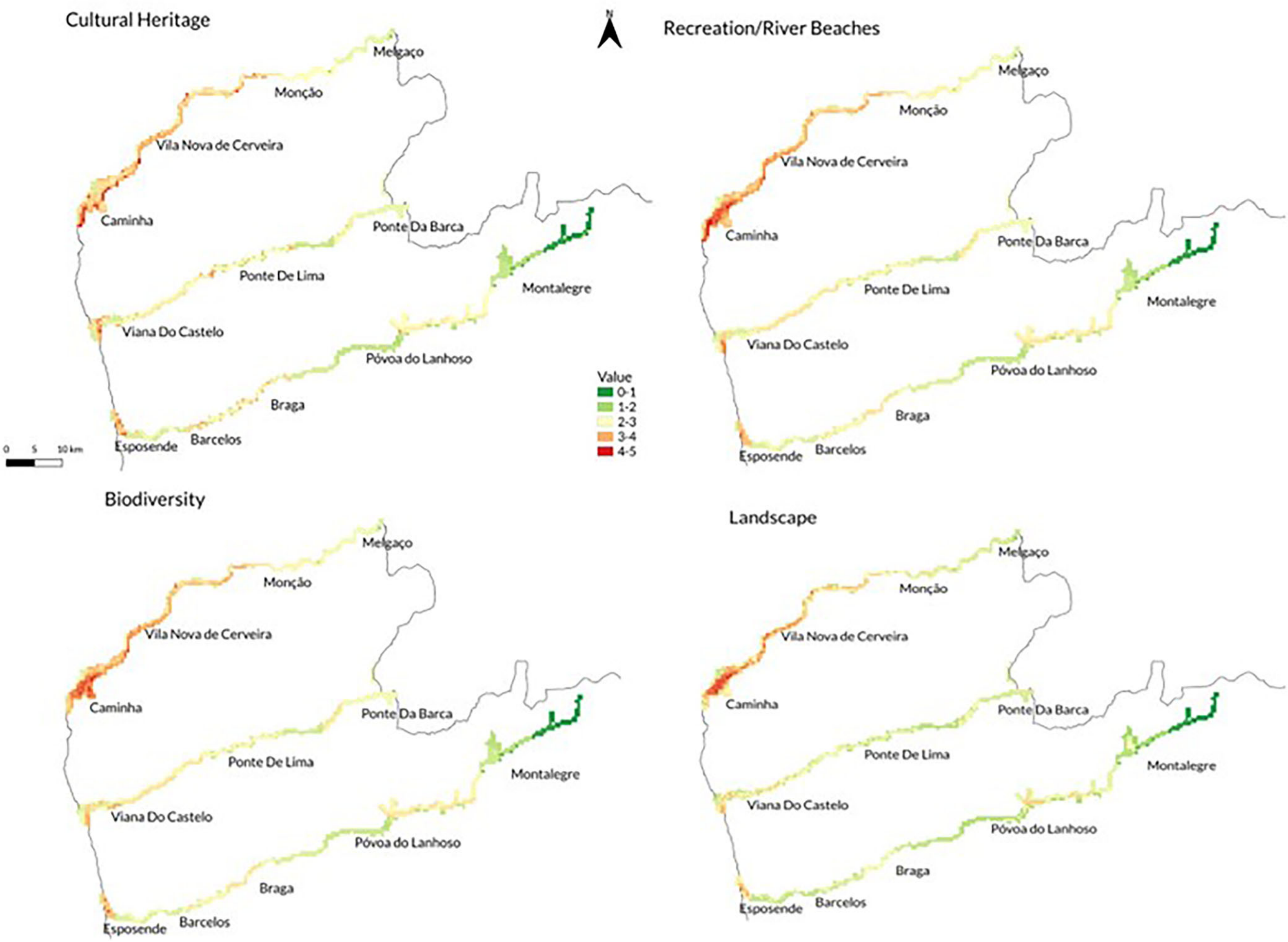

FIGURE 5 Spatial projection of cultural ecosystem services (CES) for each category. The red represents the places with the highest potential for CES provision and the dark green the places with the lowest potential according to the selected spatial variables.

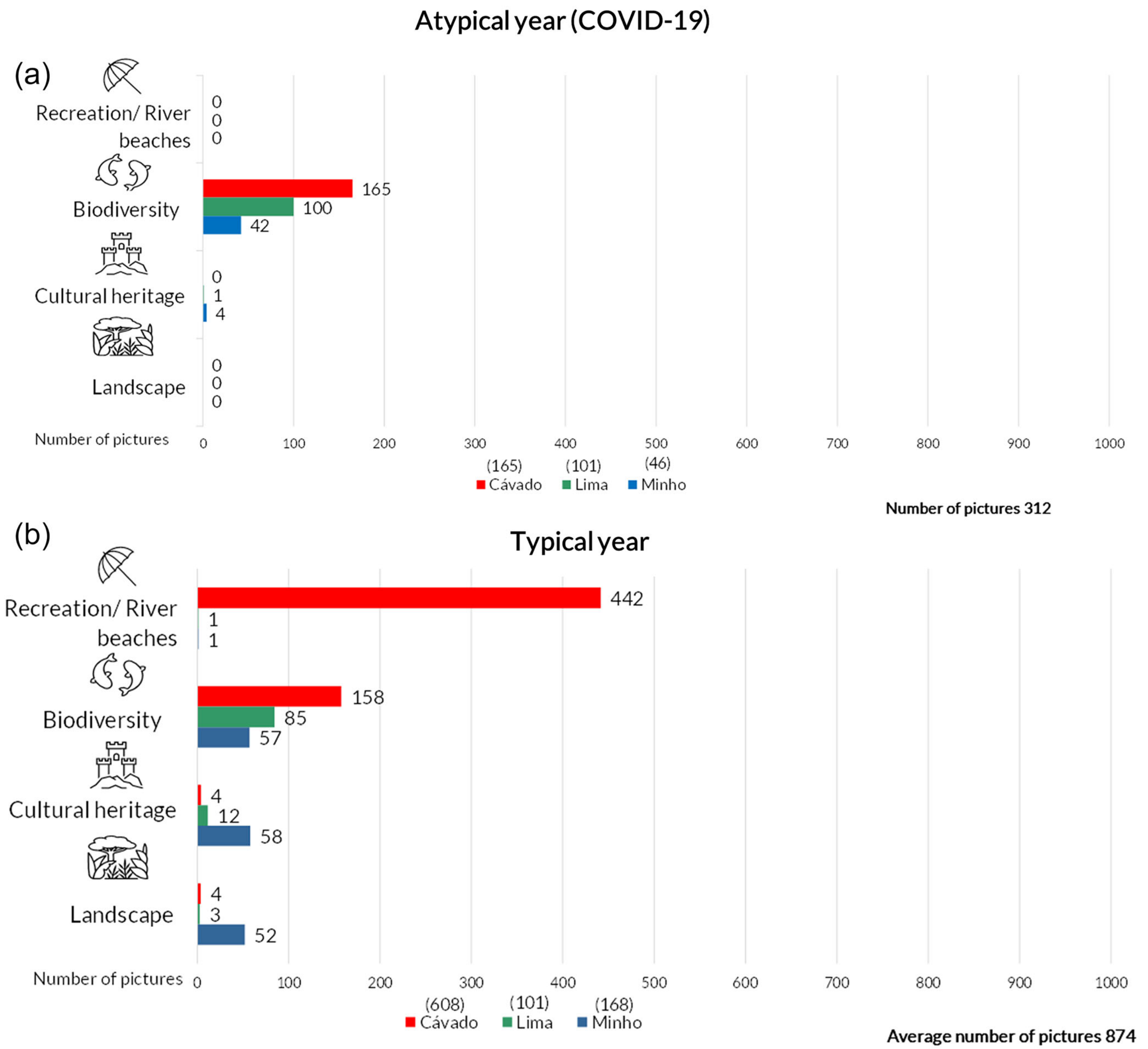

FIGURE 6 (a) Number of photographs from Flickr and iNaturalist for each category of cultural services on the Cávado, Lima and Minho rivers in the period of more restrictive measures during the COVID-19 pandemic; (b) 2018 and 2019 annual average for comparison purposes.

On the other hand, cultural heritage does not show any CES hotspot. This can be explained by the extremely low and unrepresentative number of pictures collected on social media (Flickr). As only 5 out of a total of 312 photographs were classified in this category, the statistical robustness and predictive ability is extremely low. The biodiversity category was the only that maintained the same pattern in all three rivers during the pandemic most restrictive period. A possible reason was the demand for biodiversity and nature had increased during an atypical year marked by stringent social restrictions. Indeed, some authors concur that the demand for Nature increased during the year of mandatory confinement due to COVID-19 (Beckmann-Wübbelt et al., 2021; Derks et al., 2020). One reason for this is that wildlife observation/contemplation is frequently a solitary activity and in many cases must be very quiet and undisturbed for the living creatures to be captured by the cameras and thus respecting the constraints imposed by the pandemic of social distancing. It is also worth noting that no photos were shared in the 'recreation/river beaches' and 'landscape' categories during this period, which may indicate that those who did these activities in the previous years were afraid to do them or retracted from sharing the photos on social networks for fear of social sanctions or authorities during mandatory lockdown. Regarding the CES predictors (Table 2), there is little change in variables in particular for biodiversity that is the only one represented.

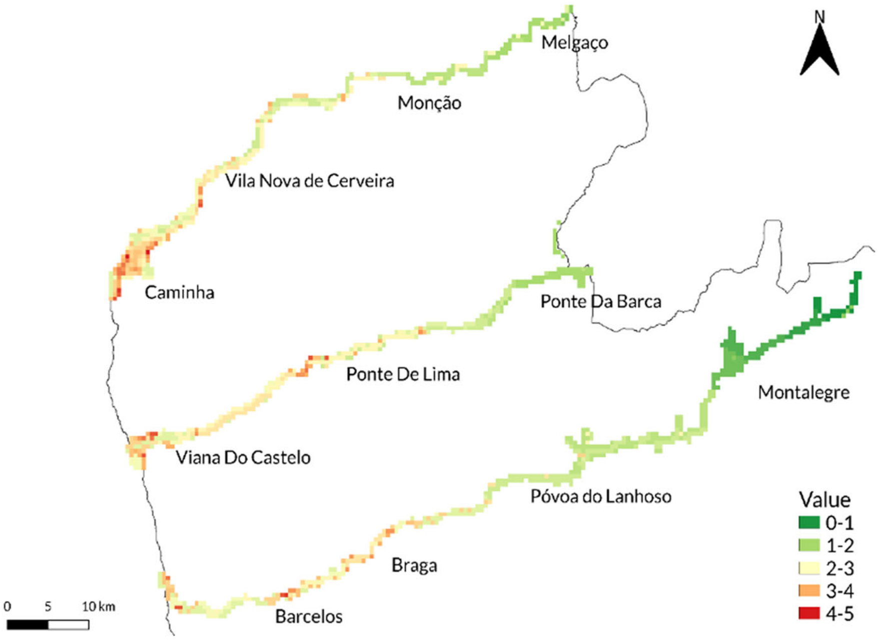

FIGURE 7 Cultural ecosystem services (CES) spatial projection of all categories for the most restrictive period during COVID-19. The red represents the places with the highest potential and the dark green the places with the lowest potential according to the selected spatial variables.

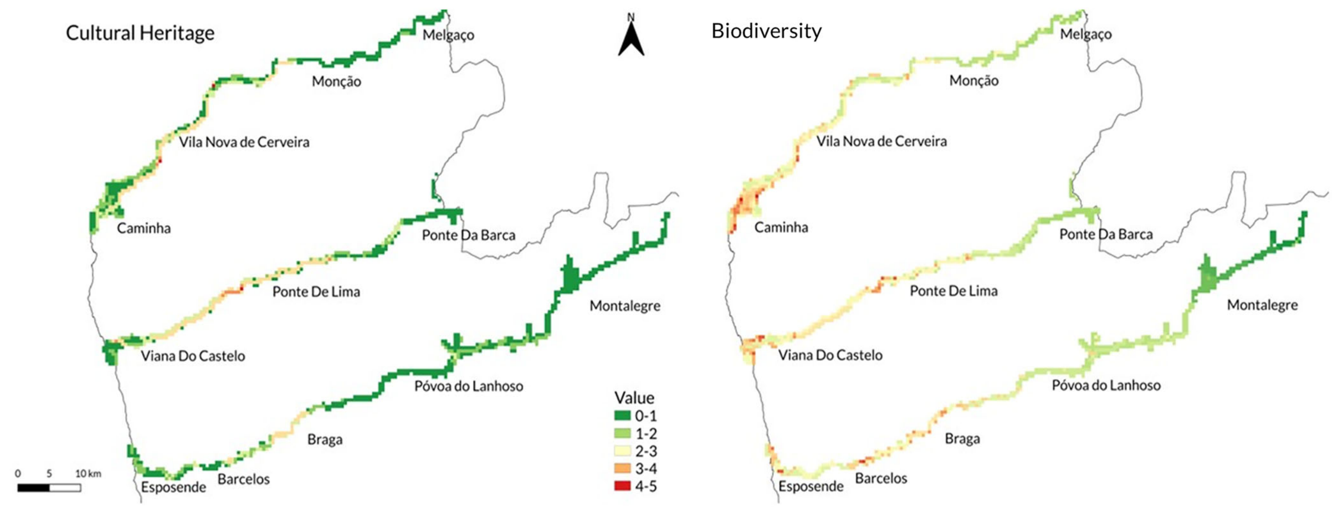

FIGURE 8 Spatial projection of cultural ecosystem services (CES) for each category for the most restrictive period during COVID-19. The red represents the places with the highest potential and the dark green the places with the lowest potential according to the selected spatial variables.

3.5 | Critical analysis of the proposed methodology

One significant benefit of this approach is its applicability to evaluating CES associated with freshwater and rivers worldwide. These areas have received less attention in scientific literature on CES mapping despite their considerable importance. The methodology employs social networks data as response variables within a biophysical context with predictive variables (Hale et al., 2019; Thiele et al., 2019; Thiele et al., 2020; Tulek, 2023). Using two complementary social platforms, a social media and a citizen science, reveals as particularly important to represent these two visions on CES uptake, one with a more natural vision and the other with a recreational vision. In addition, the use of two social media platforms may reduce uncertainty on sample size representativeness. An inherent limitation from this method lies in the absence of automation in image categorization. Due to the vast amount of data generated by social media, some researchers suggest using artificial intelligence-based photo classification to expedite this process (Cardoso et al., 2022; Mouttaki et al., 2022).

Another potential constraint pertains to the sample size, which might be skewed due to the privacy settings of platforms like Flickr, where users can prevent photo downloads. This could compromise the effectiveness of this approach. Moreover, the lack of socio-economic information about social media users hampers thorough socio-economic analyses. Integrating data about users' local or visitor status, travel distance, gender, age and profession could significantly enhance result interpretation (Cao et al., 2022). Understanding users' motives for posting photos (Dorwart et al., 2009) could also provide valuable insights. Spatial variables are crucial to ensuring result accuracy (Crouzat et al., 2022; Fonseca & Santos, 2018). Creating an extensive, open database from this social media data has the potential to consolidate information on CES and biodiversity across diverse locations, streamlining conservation prioritization and site management. However, a substantial challenge lies in the costs of data storage and maintenance, which would need commitment from custodians of the dataset.

It is our belief that studying CES solely through social networks should not occur in isolation but should be complemented by questionnaires and participatory workshops (Vieira et al., 2018). This comprehensive approach ensures a more accurate understanding of CES value by combining various data sources like on-site interviews, decision-maker workshops and online surveys.

From another perspective, our study did not account for the tributaries of the three main rivers under examination, representing a spatial limitation. Despite their smaller size, these tributaries hold significant importance for individuals seeking CES related to rivers, especially those living or conducting activities in these areas. Future studies incorporating tributary assessments can provide a more comprehensive understanding of CES demand and value for stakeholders and decision-makers.

From the methodological perspective, we believe our best model can be considered robust. First, we tested the models with a different temporal subset of pictures as dependent variable, focusing on a period of low demand for cultural services and fewer social media posts (COVID-19 2020–2021). Despite this, the explanatory power of the environmental variables produced the same mapping results as in normal years. Secondly, although in our picture collection period (2018–2019) more than half of the pictures were taken at the same location (a river beach), the final predictive model identified hotspots for CES provision in a completely different location (in river Minho estuary) demonstrating the high predictive power of the multimodel interference technique. Moreover, our study not only aims to build predictive models for CES provided by the rivers but also provides relevant insights into how a large and suitable asset of environmental variables can be used to identify hotspot areas within rivers for CES provision. Future efforts should test the predictive capacity of our variables over larger spatio-temporal scales to reinforce the empirical support of our findings.

3.6 | Opportunities for better CES territorial management and future research

Riparian forests and protected areas play a crucial role in delivering CES associated with rivers. This underscores the importance of restoring landscapes along riverbanks and how such restoration can enhance the provision of CES linked to rivers. Of particular significance is the potential value of biodiversity photographs in addressing biodiversity conservation challenges. For example, these photos can assist in identifying invasive species and monitoring their spatial and temporal spread in these regions (Koen & Newton, 2021). Because, in this study, estuaries were the hotspot areas for CES provision and use, particular targeted measures should be considered for the maintenance of nature conservation and at the same time the use of CES related to rivers.

To enhance the management of CES related to rivers, collaborative efforts can be undertaken, such as organizing workshops or local lectures in partnership with municipalities. These events can educate citizens about the necessary precautions they can take to contribute to the conservation of freshwater ecosystems (Crouzat et al., 2022).

Furthermore, it is essential to invest in infrastructure development aimed at safeguarding biodiversity. To achieve this goal, municipalities should prioritize sustainable tourism marketing initiatives. This may include the creation of guided trails with informative signage showcasing the routes and the local fauna and flora (Elwell et al., 2020). Additionally, the installation of benches along riverbanks can create public spaces that encourage people to connect with nature. Lastly, maintaining an active presence on social media and highlighting cultural aspects like local cuisine can be highly appealing to visitors (Vaz et al., 2018). These efforts can further foster an appreciation for CES and promote initiatives focused on biodiversity conservation.

On the other side, it is essential for scientific organizations, universities and educational institutions to increase their investment in researching this subject. This will enable them to gain a deeper understanding of how these natural 'refuges' (Elwell et al., 2020) can be ecologically enhanced and the implications this has for the high demand for CES associated with rivers. To achieve this, collaborative partnerships with municipalities should be nurtured, fostering continuous and open communication with local communities. By actively seeking feedback from citizens regarding new investments and ideas for better connecting them to CES on a local scale, we can promote a more inclusive and participatory approach. This community engagement can then be integrated with territorial options at the regional level, contributing to the development of an effective framework for managing CES at the territorial level (Martinez-Harms et al., 2018).

By combining scientific research, community engagement and responsive governance, we can promote the sustainable conservation and utilization of CES while simultaneously addressing the needs of local communities and safeguarding these invaluable natural resources for future generations.

4 | CONCLUSION

With the use of the methodology outlined in this study, we successfully presented significant findings regarding the potential mapping of CES associated with rivers. This approach incorporated photographs from social media and citizen science platforms, considering two different periods (typical years and an atypical year during COVID-19 most restrictive period), spatial data, multimodel inference and ultimately generated potential mapping.

An analysis of the number of photographs reveals that each river exhibits distinctive and more pronounced characteristics for CES related to rivers use, highlighting the importance of local studies applying the stepwise methodology described in this study. The Minho River exhibits a higher prevalence of photographs related to the 'historical heritage' category, whereas the Lima River is associated more with 'biodiversity', and the Cávado River is linked with 'recreation/river beaches'. This underscores the importance of all categories of CES in comprehending and conserving ecosystems, especially when considering their interactions with visitors and local populations.

Furthermore, the variables that most effectively explained the distribution of photographs included factors such as the presence of protected areas, the distance from urban areas to river beaches and tourist attractions, the number of trails, recreational and tourism points and the average annual temperature. The mapping exercise identified river estuaries as major hotspots for CES provision, underscoring their ecological significance in supporting various CES.

It is worth noting that when testing the methodology during the most restrictive period of the COVID-19 pandemic, it was observed that only patterns related to 'biodiversity' were consistently maintained in both periods. Importantly, this did not significantly influence the relationship with environmental variables and the subsequent potential mapping.

Through thoughtful interpretation, our findings hold the promise of aiding researchers and environmental managers in gaining deeper insights into human–nature interactions. They shed light on the values and benefits associated with CES related to rivers. This comprehension, in turn, can be instrumental in shaping strategies for more effective watershed management. The development and implementation of actions at the local level, geared towards promoting the responsible and sustainable utilization of CES related to rivers, emerge as imperative steps in facilitating ecosystems' adaptation to global changes.

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CONFLICT OF INTEREST STATEMENT

The authors declare no competing interests.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analysed in this study.

ETHICAL STATEMENT

Data protected by social media users' rights were not considered throughout this study, and public data containing sensitive and personal information (including images with recognizable faces or structures) were treated as confidential or (pseudo-)anonymized. Before being used for scientific presentations or other public events, images containing private data were anonymized. The user's personal information, such as name, ID (identification), age and location, was not extracted or considered.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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